Lecture 27: Eigenvalue Problems; Unitary Similarity Transformations

Math 6610

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Today's Goals

- Eigenvalues, eigenvectors, spectrum, characteristic polynomial
- Bad and better ideas for computing eigenvalues
- Similarity transformations; unitary matrices
- The Schur theorem and its proof

Eigenvalue and Eigenvector

Definition (*Eigenvalue, Eigenvector*). Let $A \in \mathbb{C}^{n\times n}$. A number $\lambda \in \mathbb{C}$ is called an *eigenvalue* of A if there exists a nonzero $x \in \mathbb{C}^n$, $x \neq 0$, such that

$$Ax = \lambda x.$$

Such an x is called an *eigenvector* of A.

Spectrum

Definition (*Spectrum*). The set

$$\Lambda(A) := \left\{ \lambda \in \mathbb{C} \;\middle|\; \lambda \text{ is an eigenvalue of } A \right\}$$

is called the *spectrum* of A.

$$\begin{aligned} & Ax = \lambda x, && x \neq 0 \\ \Rightarrow \quad & (\lambda I - A)x = 0, && x \neq 0 \\ \Rightarrow \quad & \text{the matrix } \lambda I - A \text{ is singular} \\ \Rightarrow \quad & \det(\lambda I - A) = 0 \end{aligned}$$

Characteristic Polynomial

Definition (*Characteristic Polynomial*). The polynomial

$$p(z) = p_A(z) := \det(zI - A) = z^n + \alpha_{n-1}z^{n-1} + \cdots + \alpha_1 z + \alpha_0$$

is called the *characteristic polynomial* of A.

Eigenvalues as Zeros of p_A

Remark. The eigenvalues of A are the zeros of the characteristic polynomial, p_A. In particular, $A \in \mathbb{C}^{n\times n}$ has n eigenvalues, but they are not necessarily distinct.

Remark. For $n \geqslant 5$, any algorithm for computing eigenvalues of general $A \in \mathbb{C}^{n\times n}$ has to be iterative!

- Even if this was not true, working with polynomials is not great.

Every Polynomial Is Some p_A

Any polynomial $p(z) = z^n + \alpha_{n-1}z^{n-1} + \alpha_{n-2}z^{n-2} + \cdots + \alpha_1 z + \alpha_0$ is the characteristic polynomial of a matrix $A \in \mathbb{C}^{n\times n}$. For example,

$$A = \begin{bmatrix} 0 & 0 & \cdots & 0 & -\alpha_0 \\ 1 & \ddots & \ddots & \vdots & -\alpha_1 \\ 0 & \ddots & \ddots & 0 & \vdots \\ \vdots & \ddots & \ddots & 0 & -\alpha_{n-2} \\ 0 & \cdots & 0 & 1 & -\alpha_{n-1} \end{bmatrix}$$

The Characteristic Polynomial of This A

This A has the characteristic polynomial

$$\det(zI-A) = \det \begin{bmatrix} z & & & & \alpha_0 \\ -1 & z & & & \alpha_1 \\ & -1 & \ddots & & \vdots \\ & & \ddots & z & \alpha_{n-2} \\ 0 & & & -1 & z+\alpha_{n-1} \end{bmatrix} = z^n + \alpha_{n-1} z^{n-1} + \cdots + \alpha_1 z + \alpha_0.$$

Triangular Matrices

For some classes of matrices, the eigenvalues can be found trivially. For example, an upper triangular matrix:

$$A = \begin{bmatrix} a_{11} & * & \cdots & * \\ & a_{22} & \ddots & \vdots \\ & & \ddots & * \\ 0 & & & a_{nn} \end{bmatrix}, \qquad \Lambda(A) = \{a_{11}, a_{22}, \ldots, a_{nn}\}$$

Computation of Eigenvalues: Bad Ideas

1. Forming p_A and computing $\lambda \in \Lambda(A)$ as zeros of p_A.
2. Attempting to compute the *Jordan canonical form* of A:

$$X^{-1}AX = \begin{bmatrix} J_1 & & & 0 \\ & J_2 & & \\ & & \ddots & \\ 0 & & & J_l \end{bmatrix},$$

Computation of Eigenvalues: Bad Ideas

where each

$$J_i = \begin{bmatrix} \lambda_i & 1 & & & 0 \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ 0 & & & & \lambda_i \end{bmatrix}, \qquad i = 1, 2, \ldots, l$$

are *Jordan blocks*.

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[X,J] = jordan(A)
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Computation of Eigenvalues: Better Ideas

1. Use the *Schur factorization* of A.
2. Use unitary *similarity transformations* to first transform A to a "simpler form".

Definition (*Similarity Transformations*). Let $A \in \mathbb{C}^{n\times n}$. Let $U \in \mathbb{C}^{n\times n}$ be nonsingular. The map

$$A \mapsto U^{-1}AU$$

is called a *similarity transformation* of A.

Similarity Keeps the Spectrum

Lemma.

$$\Lambda(A) = \Lambda(U^{-1}AU)$$

Proof.

$$\begin{aligned} Ax &= \lambda x, && x \neq 0 \\ U^{-1}AU(\underbrace{U^{-1}x}_{\tilde{x}}) &= \lambda(\underbrace{U^{-1}x}_{\tilde{x}}), && \tilde{x} \neq 0 \\ U^{-1}AU\tilde{x} &= \lambda\tilde{x}, && \tilde{x} \neq 0 \end{aligned}$$

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Unitary Matrices

Use matrices U that are easy to invert and that are numerically well-behaved:

Definition (*Unitary*). $U \in \mathbb{C}^{n\times n}$ is said to be *unitary* if

$$U^H U = I$$

($U = [u_{jk}]$, $U^H = [\overline{u_{kj}}] = (\overline{U})^T$.)

Properties of Unitary Matrices

1. U unitary $\Rightarrow U$ is nonsingular and $U^{-1} = U^H$.
2. U unitary $\Rightarrow U^H$ is unitary and $(U^H)^{-1} = (U^H)^H = U$.
3. U unitary $\Rightarrow \|Ux\|_2 = \|x\|_2$ for all $x \in \mathbb{C}^n$. This is because $\|Ux\|_2^2 = (Ux)^H Ux = x^H U^H Ux = x^H x = \|x\|_2^2$ and $\|U\|_2 = \max\limits_{x \neq 0} \frac{\|Ux\|_2}{\|x\|_2} = 1$.
4. Unitary matrices U have the best possible Euclidean condition numbers:

$$\kappa_2(U) = \underbrace{\|U\|_2}_{=1} \underbrace{\left\|\overbrace{U^{-1}}^{=U^H}\right\|_2}_{=1} = 1.$$

Unitary Similarity Transformations

$$A \mapsto U^{-1}AU = U^{H}AU =: T$$

where U is unitary.

Question: What is the "simplest" T we can achieve?

The Schur Theorem

Theorem (*Schur*). For any $A \in \mathbb{C}^{n\times n}$, there exists a unitary $U \in \mathbb{C}^{n\times n}$ such that

$$U^H A U = T = \begin{bmatrix} t_{11} & * & \cdots & * \\ & t_{22} & \ddots & \vdots \\ & & \ddots & * \\ 0 & & & t_{nn} \end{bmatrix}$$

is upper-triangular. In particular, $\Lambda(A) = \Lambda(T) = \{t_{11}, t_{22}, \ldots, t_{nn}\}$.

We will prove this by induction on n. For $n = 1$: $A = [a]$, $U = [1]$, $T = [a]$.

Proof of Schur's Theorem

Recall:

$$U^H A U = T = \text{upper-triangular}$$

Proof. By induction. For $n = 1$, it is trivial. Assume it is true for $1, \ldots, n-1$. Let $\lambda \in \Lambda(A)$ with eigenvector x, $\|x\|_2 = 1$. Choose a unitary matrix

$$U_1 = \begin{bmatrix} x & * & * & \cdots & * \end{bmatrix} \in \mathbb{C}^{n \times n}$$

with x as the first column. Then:

$$AU_1 = \begin{bmatrix} \lambda x & * & * & \cdots & * \end{bmatrix}$$

$$U_1^H A U_1 = \begin{bmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & \tilde{A} & \\ 0 & & & \end{bmatrix}, \qquad \text{where } \tilde{A} \in \mathbb{C}^{(n-1)\times(n-1)}$$

(continued on the next slide)

Proof of Schur's Theorem

Induction hypothesis:

$$\tilde{U}^H \tilde{A} \tilde{U} = \tilde{T} = \begin{bmatrix} * & * & \cdots & * \\ & * & \cdots & * \\ & & \ddots & \vdots \\ 0 & & & * \end{bmatrix}, \qquad \tilde{U} \in \mathbb{C}^{(n-1)\times(n-1)} \text{ is unitary}$$

Set

$$U := U_1 \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \tilde{U} & \\ 0 & & & \end{bmatrix} \in \mathbb{C}^{n\times n}$$

(continued on the next slide)

Proof of Schur's Theorem

Proof (continued). Then U is unitary and

$$U^H A U = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \tilde{U}^H & \\ 0 & & & \end{bmatrix} U_1^H A U_1 \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \tilde{U} & \\ 0 & & & \end{bmatrix}$$

$$= \begin{bmatrix} \lambda & * & \cdots & * \\ 0 & & & \\ \vdots & & \tilde{U}^H \tilde{A} \tilde{U} & \\ 0 & & & \end{bmatrix} =: T = \text{upper-triangular}$$

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Looking Ahead

- **Next lecture:** Householder Reflectors; Reduction to Hessenberg Form